

STAAR 2.0 READY

STAAR Science Task Cards

Grade 5

32 TEKS-Aligned Practice Questions

- ★ STAAR 2.0 format practice
- ★ Visual aids & diagrams
- ★ Answer key included
- ★ Recording sheet

Teach4Texas

Professional Resources for Texas Educators



1**TEKS 5.5A**

A student measures the mass and volume of four different liquids. Which property is BEST used to identify an unknown liquid?

Liquid	Mass (g)	Volume (mL)	Density (g/mL)
Water	100	100	1.00
Vegetable Oil	92	100	0.92
Corn Syrup	138	100	1.38
Unknown	79	100	0.79

- A)** Color
B) Density
C) Temperature
D) Shape of the container

2**TEKS 5.5A**

A student has a piece of aluminum foil, a wooden block, and a glass marble. Which physical property do all three objects share?

- A)** They are all the same color
B) They all conduct electricity
C) They all have mass and take up space
D) They are all magnetic

3**TEKS 5.5B**

A student mixes sand, iron filings, and salt together in a bowl. Which method would BEST separate the iron filings from the mixture first?

- A)** Add water and stir
B) Use a magnet
C) Pour through a filter
D) Heat the mixture

4**TEKS 5.5B**

A student dissolves sugar in warm water and stirs. Which statement BEST describes this mixture?

- A)** A new substance has been formed
B) The sugar has been destroyed
C) The sugar can be recovered by evaporating the water
D) The sugar changed into a different chemical

5**TEKS 5.5C**

Students recorded the temperature of water as it was heated. Based on the data, at what temperature did the water begin to boil?

Time (min)	Temperature (°C)	Observations
0	22	No change
2	45	Small bubbles forming
4	70	More bubbles
6	90	Large bubbles
8	100	Rapid bubbling, steam rising
10	100	Still boiling, steam

- A)** 80°C
B) 90°C
C) 100°C
D) 110°C

6**TEKS 5.5C**

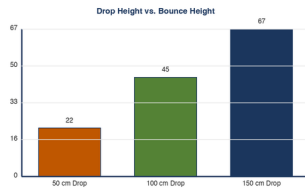
Ice cream is left on a kitchen counter on a hot day. Which change of state occurs as the ice cream melts?

- A)** Gas to liquid
B) Liquid to solid
C) Solid to liquid
D) Liquid to gas

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TEKS 5.6A

A student drops a ball from three different heights and measures the bounce height each time. Based on the bar graph, what conclusion can the student make?



- A) The ball bounces lower when dropped from a greater height
 B) The ball bounces higher when dropped from a greater height
 C) Drop height does not affect bounce height
 D) The ball always bounces to the same height

8

TEKS 5.6A

Two students push a wagon. Student A pushes with a force of 10 N to the right, and Student B pushes with a force of 15 N to the right. What is the net force on the wagon?



- A) 5 N to the right
 B) 10 N to the right
 C) 15 N to the right
 D) 25 N to the right

9

TEKS 5.6A

A book is sitting on a table and not moving. What can you conclude about the forces acting on the book?



- A) No forces are acting on the book
 B) The forces are balanced
 C) Gravity is not acting on the book
 D) The forces are unbalanced

10

TEKS 5.6B

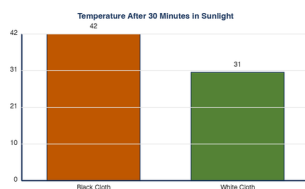
Which of the following is the BEST example of light energy being transformed into heat energy?

- A) A flashlight shining on a wall
 B) Sunlight warming a dark sidewalk
 C) A mirror reflecting a light beam
 D) A lamp lighting up a room

11

TEKS 5.6B

A student wraps a thermometer in black cloth and another in white cloth, then places both in sunlight for 30 minutes. The bar graph shows the results. Why did the black cloth reach a higher temperature?



- A) Black cloth reflects more light
 B) Black cloth absorbs more light energy and converts it to heat

12

TEKS 5.6C

A guitar string vibrates when plucked. What happens to the sound if the string vibrates faster?

- A) The sound gets louder
 B) The sound gets softer

C) White cloth produces more heat

D) Black cloth is always hotter than white cloth

C) The pitch gets higher

D) The pitch gets lower

13

TEKS 5.7A

Which step of the water cycle is occurring when water vapor in the atmosphere cools and forms clouds?

- A) Evaporation B) Condensation
C) Precipitation D) Collection

14

TEKS 5.7A

A student sets up an experiment with a covered bowl of warm water in sunlight. After 2 hours, water droplets form on the inside of the cover. Which TWO parts of the water cycle does this experiment demonstrate?

- A) Precipitation and collection B) Evaporation and condensation
C) Condensation and precipitation D) Collection and evaporation

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TEKS 5.7B

A weather station recorded data for five days. Based on the table, on which day is a thunderstorm MOST likely?

Day	Temperature (°F)	Humidity (%)	Air Pressure (mb)	Cloud Cover
Monday	72	35	1018	Clear
Tuesday	75	50	1015	Partly Cloudy
Wednesday	80	65	1010	Mostly Cloudy
Thursday	85	90	1002	Overcast
Friday	70	40	1020	Clear

- A) Monday B) Tuesday
C) Thursday D) Friday

16

TEKS 5.7C

A scientist finds a fossil of a fish in rock layers high on a mountain. What does this fossil evidence BEST suggest about this area?

- A) Fish once lived on mountains B) The area was once covered by water
C) The mountain formed underwater recently D) Someone placed the fish fossil there

17

TEKS 5.7C

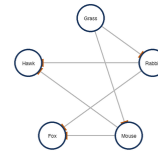
A student examines four rock samples. Which process is responsible for forming sedimentary rock?

- A) Melting and cooling of magma B) Extreme heat and pressure underground
C) Layers of sediment compacting and cementing over time D) Volcanic eruptions depositing lava

18

TEKS 5.8A

In the food web shown, what would MOST LIKELY happen to the population of rabbits if the foxes were removed from the ecosystem?

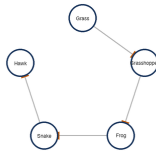


- A) The rabbit population would decrease B) The rabbit population would increase
C) The rabbit population would stay the same D) The rabbits would leave the ecosystem

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TEKS 5.8A

Look at the food web. Which organism is a producer?



- A) Grasshopper B) Frog
C) Grass D) Snake

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TEKS 5.8A

In a pond ecosystem, algae use sunlight to make food. Small fish eat the algae. Larger fish eat the small fish. Which organism in this food chain is a primary consumer?

- A) The sun B) The algae
C) The small fish D) The larger fish

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TEKS 5.8B

A cactus has thick, waxy skin and stores water in its stem. These characteristics are BEST described as adaptations for surviving in which environment?

- A) Tropical rainforest B) Desert
C) Arctic tundra D) Freshwater pond

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TEKS 5.8B

A scientist measured the beak lengths of four bird species and the types of food they eat. Which bird is BEST adapted for cracking open hard seeds?

Bird	Beak Shape	Beak Length (cm)	Main Food
Bird A	Long, thin	8.2	Nectar from flowers
Bird B	Short, thick	2.1	Hard seeds
Bird C	Flat, wide	4.5	Filtering water organisms
Bird D	Curved, hooked	3.8	Tearing meat

- A) Bird A — long, thin beak B) Bird B — short, thick beak
C) Bird C — flat, wide beak D) Bird D — curved, hooked beak

23

TEKS 5.8C

Which of the following traits is MOST LIKELY inherited from a puppy's parents?

- A) The ability to sit on command B) A scar on the puppy's ear
C) Fur color and pattern D) A preference for a certain toy

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TEKS 5.8C

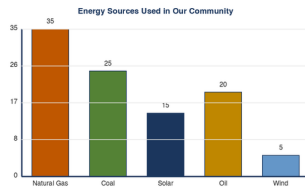
A mother plant has red flowers. All of her offspring also have red flowers. A nearby plant of the same species has white flowers. Which statement BEST explains this observation?

- A) The soil determines flower color B) Flower color is an inherited trait passed from parent to offspring
C) All plants of the same species have the same flower color D) The amount of sunlight determines flower color

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TEKS 5.9A

Students collected data on energy sources used in their community. Based on the graph, which energy source is renewable?



- A) Natural gas B) Coal
C) Solar D) Oil

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TEKS 5.9A

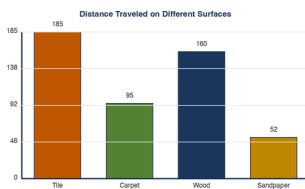
Which of the following is a nonrenewable resource?

- A) Wind B) Sunlight
C) Natural gas D) Water flowing in a river

27

TEKS 5.6A

A student tests how the surface type affects the distance a toy car travels. The results are shown in the graph. Which surface created the MOST friction?

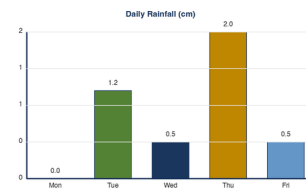


- A) Tile B) Carpet
C) Wood D) Sandpaper

28

TEKS 5.7A

Students measured rainfall in their schoolyard for one week. Based on the data, what was the total rainfall for the week?



- A) 2.0 cm B) 3.5 cm
C) 4.2 cm D) 5.0 cm

29

TEKS 5.8A

In a forest food web, a disease kills most of the mice. What will MOST LIKELY happen to the snake population that feeds on mice?



30

TEKS 5.5A

A student wants to determine whether an unknown mineral is calcite or quartz. She tests both minerals for hardness using a scratch test. Based on the data table, which conclusion is BEST supported?

Test	Calcite (hardness 3)	Quartz (hardness 7)	Unknown Mineral
Scratched by fingernail?	No	No	No
Scratched by copper penny?	Yes	No	Yes
Scratched by glass?	Yes	No	Yes
Scratches glass?	No	Yes	No

- A) The unknown mineral is calcite because it B) The unknown mineral is quartz because it

- | | |
|--|--|
| A) The snake population will increase | B) The snake population will decrease |
| C) The snake population will not change | D) The snakes will become herbivores |

- | | |
|---|--|
| scratched the glass | scratched the glass |
| C) The unknown mineral is calcite because glass scratched it | D) The unknown mineral is quartz because nothing scratched it |

31**TEKS 5.6B**

A student rubs her hands together quickly. Her hands begin to feel warm. Which energy transformation is taking place?

- A)** Heat energy to mechanical energy
- B)** Mechanical energy to heat energy
- C)** Light energy to heat energy
- D)** Chemical energy to light energy

32**TEKS 5.9A**

A town wants to reduce its use of nonrenewable resources. The data table shows their current energy usage and a proposed plan. Which change in the plan would MOST reduce their use of nonrenewable resources?

Energy Source	Current Usage (%)	Proposed Plan (%)
Coal	30	5
Natural Gas	25	20
Oil	20	5
Solar	10	35
Wind	15	35

- A)** Increasing coal usage
- B)** Replacing coal and oil with wind and solar
- C)** Using more natural gas
- D)** Reducing all energy usage to zero

Answer Key

#	TEKS	Answer	Correct Choice
1	TEKS 5.5A	B	B) Density
2	TEKS 5.5A	C	C) They all have mass and take up space
3	TEKS 5.5B	B	B) Use a magnet
4	TEKS 5.5B	C	C) The sugar can be recovered by evaporating the water
5	TEKS 5.5C	C	C) 100°C
6	TEKS 5.5C	C	C) Solid to liquid
7	TEKS 5.6A	B	B) The ball bounces higher when dropped from a greater height
8	TEKS 5.6A	D	D) 25 N to the right
9	TEKS 5.6A	B	B) The forces are balanced
10	TEKS 5.6B	B	B) Sunlight warming a dark sidewalk
11	TEKS 5.6B	B	B) Black cloth absorbs more light energy and converts it to heat
12	TEKS 5.6C	C	C) The pitch gets higher
13	TEKS 5.7A	B	B) Condensation
14	TEKS 5.7A	B	B) Evaporation and condensation
15	TEKS 5.7B	C	C) Thursday
16	TEKS 5.7C	B	B) The area was once covered by water
17	TEKS 5.7C	C	C) Layers of sediment compacting and cementing over time
18	TEKS 5.8A	B	B) The rabbit population would increase
19	TEKS 5.8A	C	C) Grass
20	TEKS 5.8A	C	C) The small fish
21	TEKS 5.8B	B	B) Desert
22	TEKS 5.8B	B	B) Bird B — short, thick beak
23	TEKS 5.8C	C	C) Fur color and pattern
24	TEKS 5.8C	B	B) Flower color is an inherited trait passed from parent to offspring
25	TEKS 5.9A	C	C) Solar
26	TEKS 5.9A	C	C) Natural gas
27	TEKS 5.6A	D	D) Sandpaper
28	TEKS 5.7A	C	C) 4.2 cm
29	TEKS 5.8A	B	B) The snake population will decrease
30	TEKS 5.5A	C	C) The unknown mineral is calcite because glass scratched it

#	TEKS	Answer	Correct Choice
31	TEKS 5.6B	B	B) Mechanical energy to heat energy
32	TEKS 5.9A	B	B) Replacing coal and oil with wind and solar

Student Recording Sheet

Name: _____ Date: _____

#1 A B C D	#2 A B C D	#3 A B C D	#4 A B C D
#5 A B C D	#6 A B C D	#7 A B C D	#8 A B C D
#9 A B C D	#10 A B C D	#11 A B C D	#12 A B C D
#13 A B C D	#14 A B C D	#15 A B C D	#16 A B C D
#17 A B C D	#18 A B C D	#19 A B C D	#20 A B C D
#21 A B C D	#22 A B C D	#23 A B C D	#24 A B C D
#25 A B C D	#26 A B C D	#27 A B C D	#28 A B C D
#29 A B C D	#30 A B C D	#31 A B C D	#32 A B C D